QBS 103 Final Project - Submission 2

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Data Import

```
# Import the three original datasets
covar <- read.csv("GSE85307_covar-1.csv")

gene_expr <- read.csv(
  "GSE85307_geneExpression.csv",
  row.names = 1,
  check.names = FALSE
)

gene_names <- read.csv("GSE85307_geneNames.csv")
```

Data Preparation

Sample Alignment

```
# Verify sample IDs before aligning the datasets
stopifnot(
  !anyDuplicated(covar$geoID),
  !anyDuplicated(colnames(gene_expr)),
  setequal(covar$geoID, colnames(gene_expr))
)

# Align expression columns with covariate sample IDs
sample_order <- match(covar$geoID, colnames(gene_expr))

gene_expr_aligned <- gene_expr[
  ,
  sample_order,
  drop = FALSE
]

# Confirm the sample order
stopifnot(
  identical(covar$geoID, colnames(gene_expr_aligned))
)
```

Covariate Cleaning

```
# Clean continuous and categorical variables
covar_clean <- covar %>%
  mutate(
    mat_age_yrs = as.numeric(
      str_remove(mat_age_yrs, ".*: ")
    ),

    mat_BMI = as.numeric(
      str_remove(mat_BMI, ".*: ")
    ),

    preg_preeclampsia = factor(
      str_remove(preg_preeclampsia, ".*: "),
      levels = c("Normal", "Preeclampsia")
    ),

    BMI_group = case_when(
      mat_BMI < 25 ~ "BMI < 25",
      mat_BMI >= 25 ~ "BMI >= 25",
      TRUE ~ NA_character_
    ),

    BMI_group = factor(
      BMI_group,
      levels = c("BMI < 25", "BMI >= 25")
    )
  )

# Check missing values in the analysis covariates
colSums(is.na(covar_clean[
  ,
  c("mat_age_yrs", "mat_BMI",
    "preg_preeclampsia", "BMI_group")
]))
```

```
##       mat_age_yrs           mat_BMI preg_preeclampsia         BMI_group
##                 0                 0                 0                 0
```

```
# Summarize sample sizes across the four groups
count(covar_clean, preg_preeclampsia, BMI_group)
```

```
##   preg_preeclampsia BMI_group  n
## 1            Normal  BMI < 25 40
## 2            Normal BMI >= 25 70
## 3      Preeclampsia  BMI < 25 13
## 4      Preeclampsia BMI >= 25 34
```

Gene Selection

```
# Identify genes with probes present in the expression matrix
available_genes <- gene_names %>%
  filter(
    as.character(PROBEID) %in% rownames(gene_expr_aligned)
  ) %>%
  distinct(PROBEID, GENE, .keep_all = TRUE)

# Select three genes for comparison
selected_genes <- c("PTX3", "ABCA9", "ACTG2")

# Find their probe IDs from the annotation data
selected_probes <- available_genes %>%
  filter(GENE %in% selected_genes)

# Verify that each selected gene has exactly one probe
probe_counts <- selected_probes %>%
  count(GENE, name = "n_probes")

stopifnot(
  setequal(probe_counts$GENE, selected_genes),
  all(probe_counts$n_probes == 1)
)

# Display selected genes and their probe IDs
selected_probes %>%
  select(GENE, PROBEID)
```

```
##    GENE PROBEID
## 1  PTX3 8083594
## 2 ABCA9 8017927
## 3 ACTG2 8042788
```

Preparing the Analysis Dataset

```
# Extract expression data using the matched probe IDs
selected_expression <- gene_expr_aligned[
  as.character(selected_probes$PROBEID),
  ,
  drop = FALSE
]

# Transpose expression data so each row represents a sample
expression_df <- as.data.frame(
  t(selected_expression)
)

# Match probe IDs to gene names
gene_order <- match(
  colnames(expression_df),
```

```
  as.character(selected_probes$PROBEID)
)

colnames(expression_df) <- selected_probes$GENE[gene_order]

# Move sample IDs into a column for merging
expression_df <- expression_df %>%
  rownames_to_column(var = "geoID")

# Merge expression values with cleaned covariates
analysis_data_2 <- covar_clean %>%
  left_join(expression_df, by = "geoID")

# Verify the merged analysis dataset
stopifnot(
  nrow(analysis_data_2) == nrow(covar_clean),
  identical(analysis_data_2$geoID, covar_clean$geoID),
  !anyDuplicated(analysis_data_2$geoID),
  all(selected_genes %in% names(analysis_data_2)),
  all(colSums(is.na(analysis_data_2[, selected_genes])) == 0)
)

# Summarize expression values for the selected genes
analysis_data_2 %>%
  select(all_of(selected_genes)) %>%
  summary()
```

```
##       PTX3            ABCA9            ACTG2
##  Min.   :4.552   Min.   :4.498   Min.   : 6.116
##  1st Qu.:4.776   1st Qu.:4.643   1st Qu.: 6.330
##  Median :4.867   Median :4.720   Median : 6.397
##  Mean   :4.884   Mean   :4.717   Mean   : 6.444
##  3rd Qu.:4.951   3rd Qu.:4.781   3rd Qu.: 6.471
##  Max.   :5.651   Max.   :4.967   Max.   :10.390
```

Reusable Plotting Function

```
# Define a reusable function for three types of gene expression plots
plot_gene_expression <- function(
    data,
    genes,
    continuous_var,
    categorical_vars
) {

  # Validate the requested variables
  if (length(genes) < 1) {
    stop("Please provide at least one gene.")
  }

  if (length(categorical_vars) != 2) {
```

```
  stop("Please provide exactly two categorical variables.")
}

required_vars <- c(genes, continuous_var, categorical_vars)

if (!all(required_vars %in% names(data))) {
  stop("One or more requested variables are missing.")
}

# Create a list to store the plots
plots <- list()

for (gene in genes) {

  # Plot 1: Gene expression distribution
  hist_plot <- ggplot(
    data,
    aes(x = .data[[gene]])
  ) +
    geom_histogram(
      bins = 15,
      fill = "#5B8DB8",
      color = "white"
    ) +
    labs(
      title = paste("Distribution of", gene, "Expression"),
      x = paste(gene, "Expression Level"),
      y = "Number of Samples"
    ) +
    theme_classic(base_size = 12)

  # Plot 2: Gene expression and continuous covariate
  scatter_plot <- ggplot(
    data,
    aes(
      x = .data[[continuous_var]],
      y = .data[[gene]]
    )
  ) +
    geom_point(
      color = "#5B8DB8",
      alpha = 0.7,
      size = 2
    ) +
    geom_smooth(
      method = "lm",
      se = TRUE,
      color = "#C46A5A",
      linewidth = 0.8
    ) +
    labs(
      title = paste(gene, "Expression vs. Maternal Age"),
      x = continuous_var,
```

```
      y = paste(gene, "Expression Level")
    ) +
    theme_classic(base_size = 12)

  # Plot 3: Gene expression by two categorical covariates
  box_plot <- ggplot(
    data,
    aes(
      x = .data[[categorical_vars[1]]],
      y = .data[[gene]],
      fill = .data[[categorical_vars[2]]]
    )
  ) +
    geom_boxplot(
      position = position_dodge(width = 0.8),
      width = 0.65,
      alpha = 0.8,
      outlier.shape = NA
    ) +
    geom_point(
      position = position_jitterdodge(
        jitter.width = 0.15,
        dodge.width = 0.8,
        seed = 123
      ),
      alpha = 0.45,
      size = 1.2
    ) +
    labs(
      title = paste(gene, "Expression by Pregnancy Outcome and BMI"),
      x = categorical_vars[1],
      y = paste(gene, "Expression Level"),
      fill = categorical_vars[2]
    ) +
    theme_classic(base_size = 12) +
    theme(
      legend.position = "top"
    )

  # Store all three plots for the current gene
  plots[[gene]] <- list(
    histogram = hist_plot,
    scatterplot = scatter_plot,
    boxplot = box_plot
  )
 }

 return(plots)
}
```

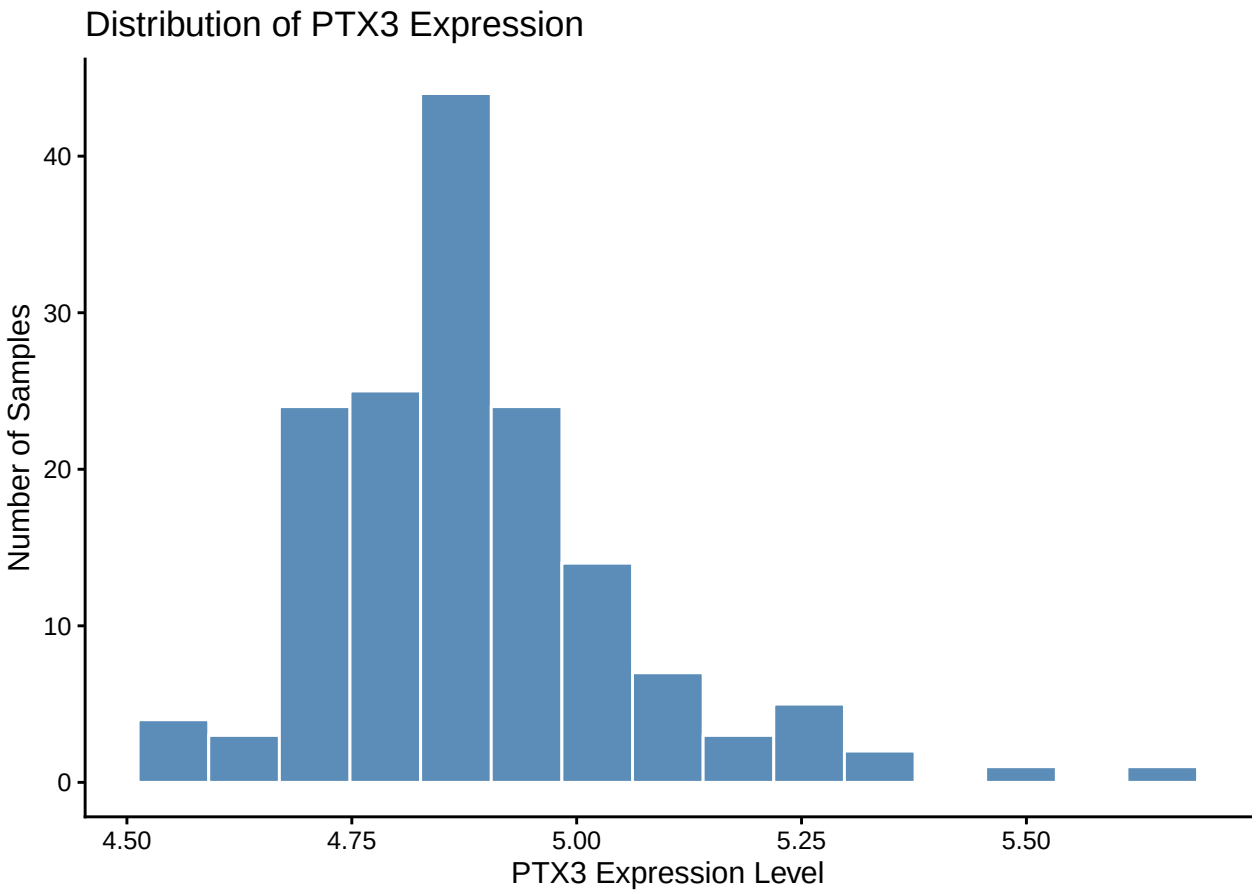
Generate Plots

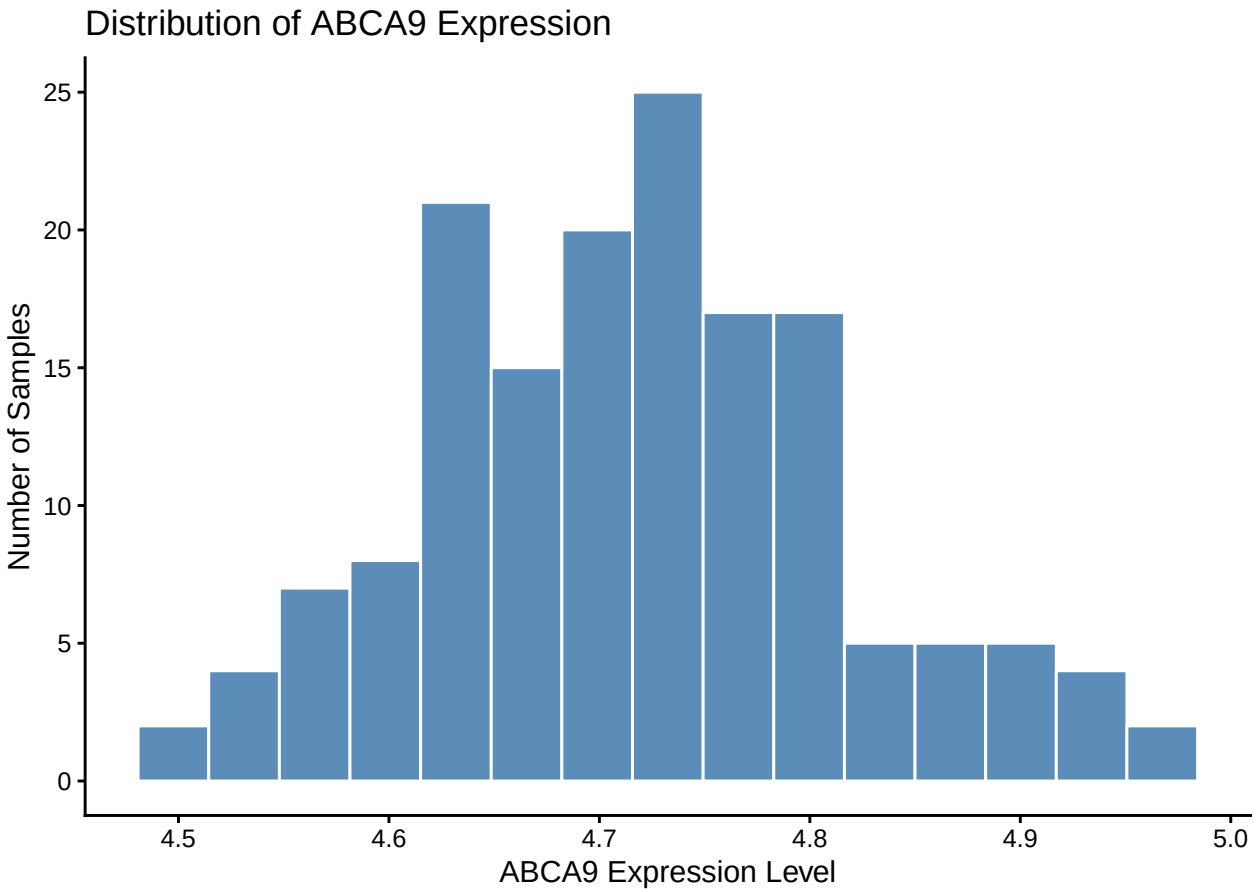
```
# Generate all nine plots with one function call
all_plots <- plot_gene_expression(
  data = analysis_data_2,
  genes = selected_genes,
  continuous_var = "mat_age_yrs",
  categorical_vars = c("preg_preeclampsia", "BMI_group")
)
```

Results

Figure Set 1: Gene Expression Distributions

```
# Display one histogram for each selected gene
for (gene in selected_genes) {
  print(all_plots[[gene]][["histogram"]])
}
```




Distribution of ABCA9 Expression
Number of Samples
0
5
10
15
20
25
4.5
4.6
4.7
4.8
4.9
5.0
ABCA9 Expression Level

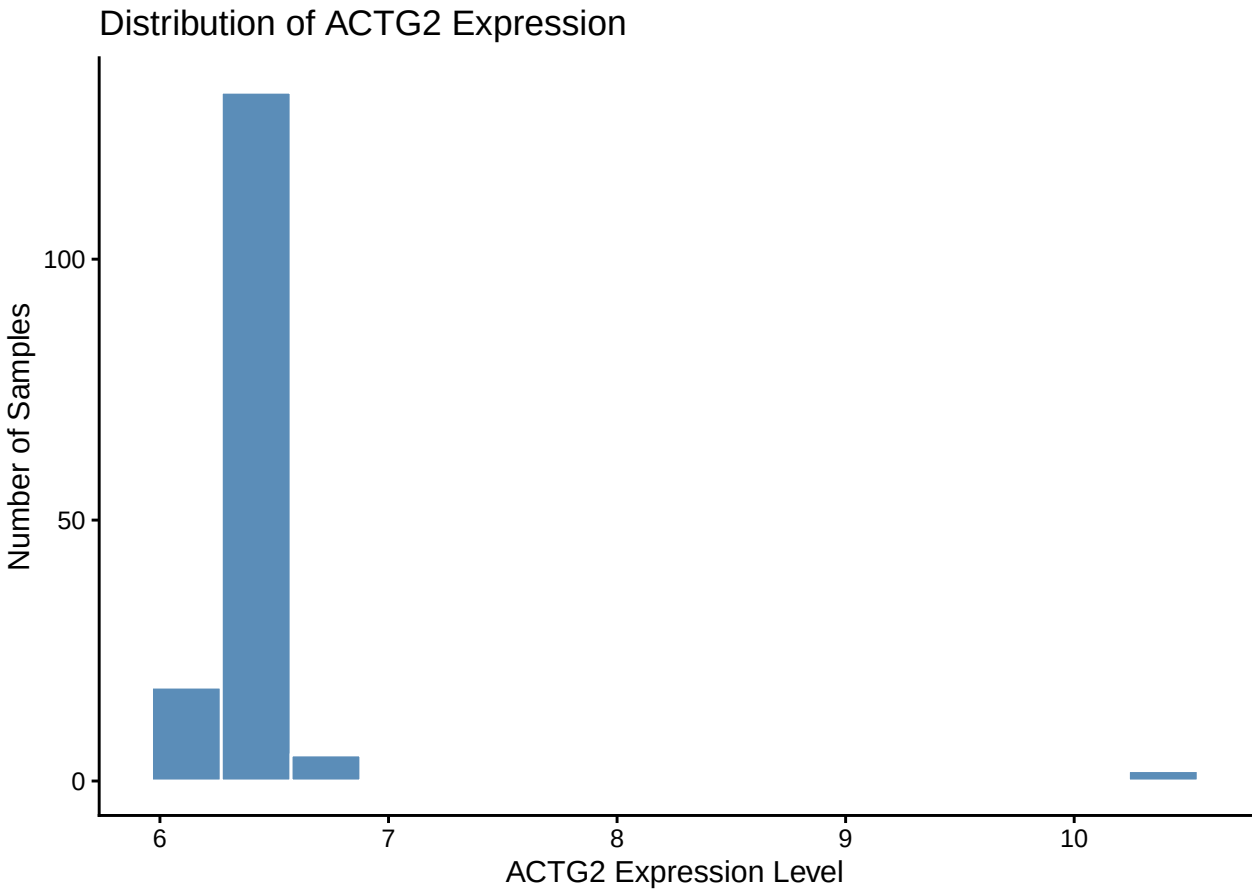
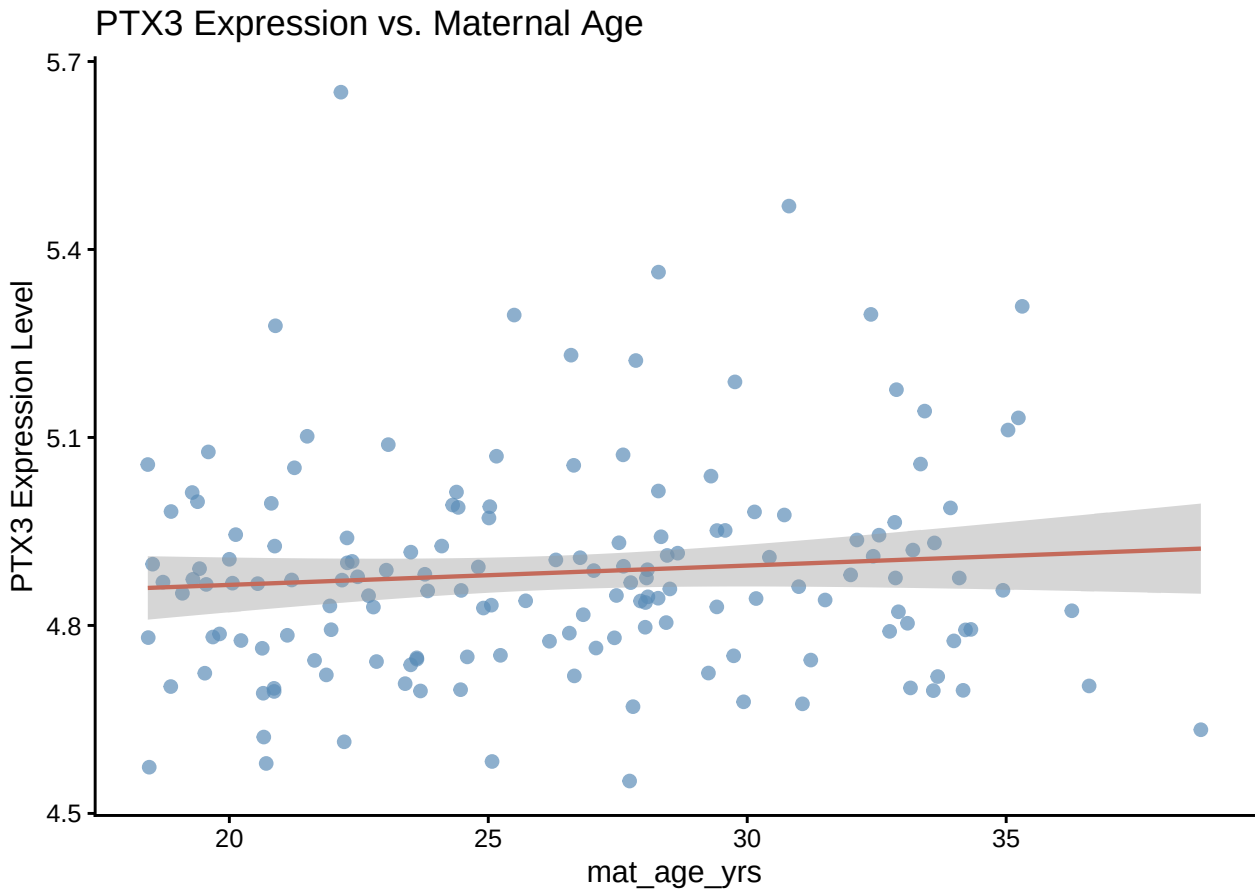
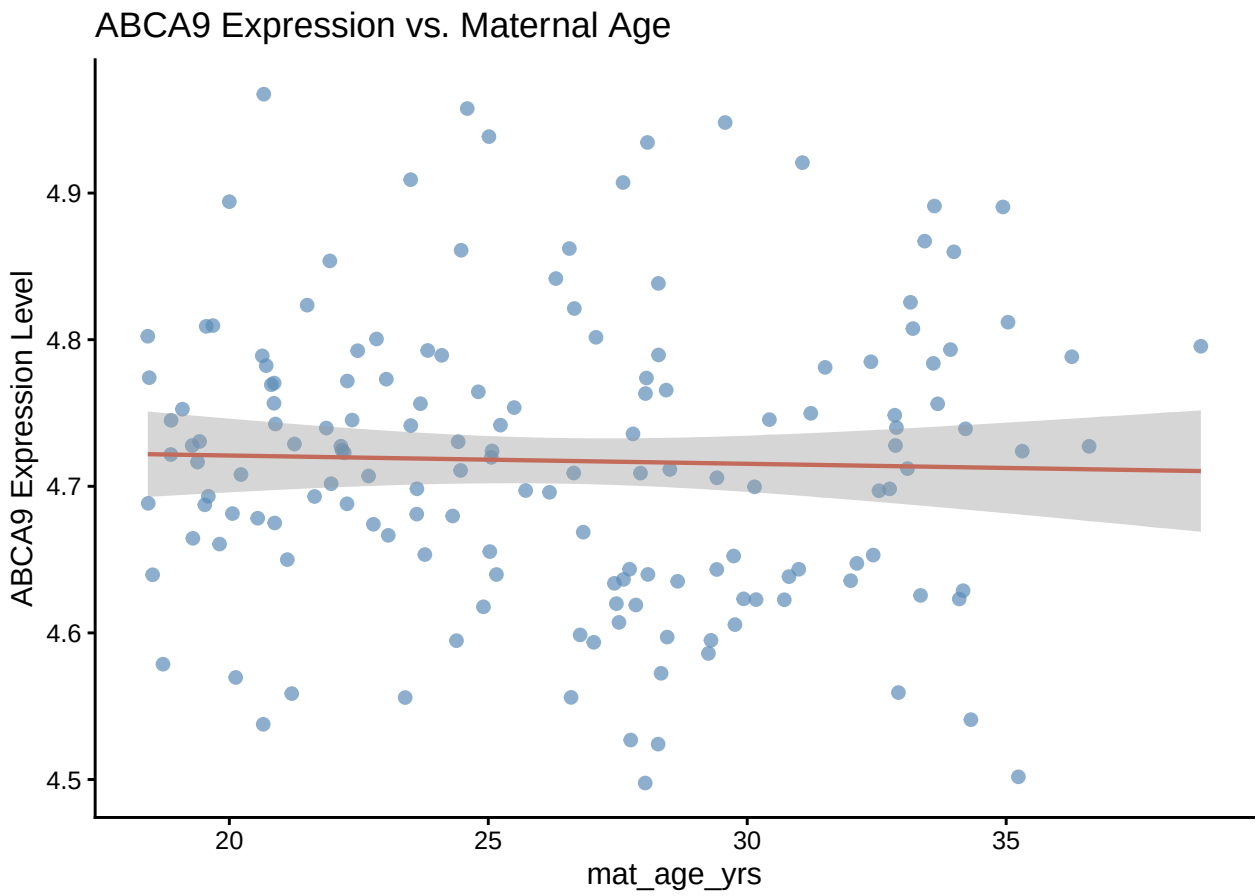


Figure Set 2: Maternal Age and Gene Expression

```
# Display one scatterplot for each selected gene
for (gene in selected_genes) {
  print(all_plots[[gene]][["scatterplot"]])
}
```


PTX3 Expression vs. Maternal Age
PTX3 Expression Level
5.7
5.4
5.1
4.8
4.5
20
25
30
35
mat_age_yrs


ABCA9 Expression vs. Maternal Age
ABCA9 Expression Level
4.9
4.8
4.7
4.6
4.5
20
25
30
35
mat_age_yrs

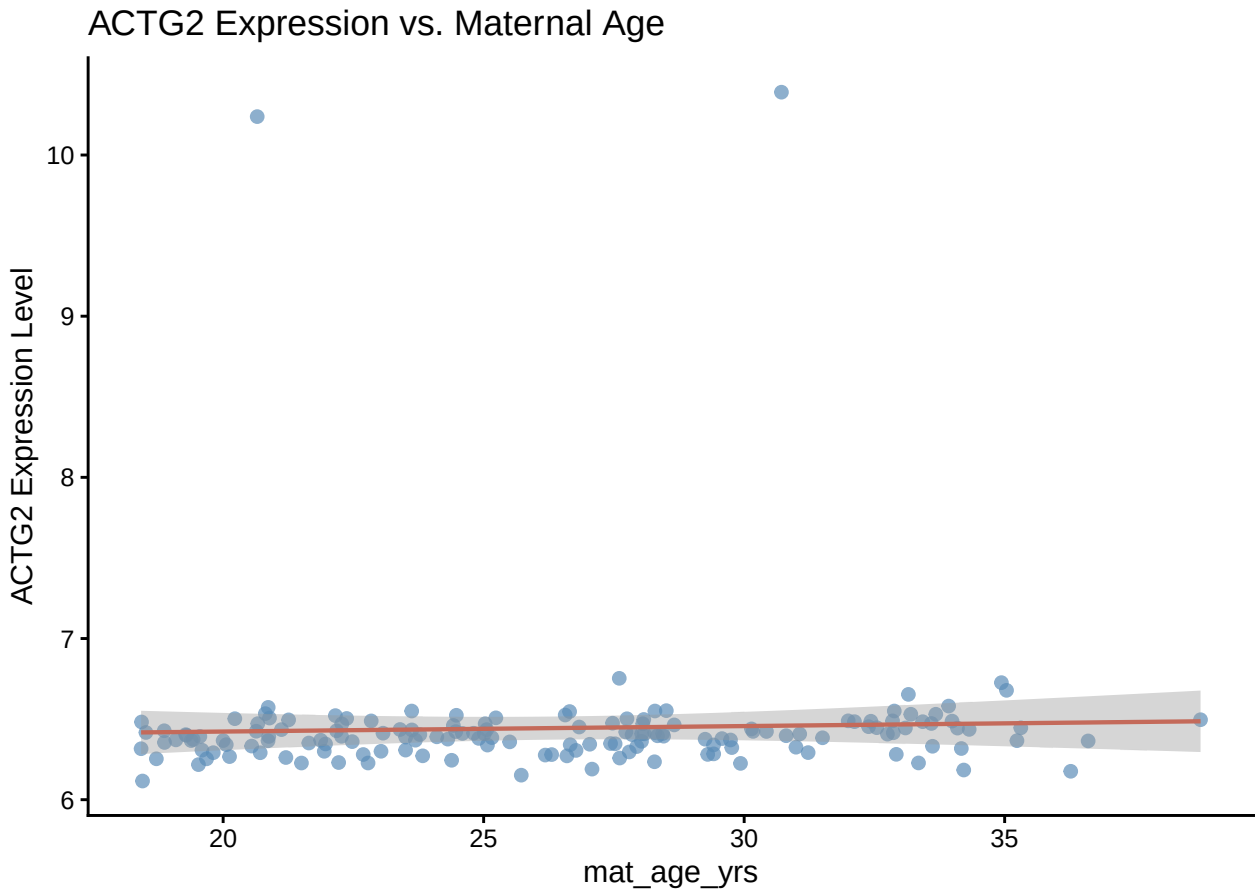
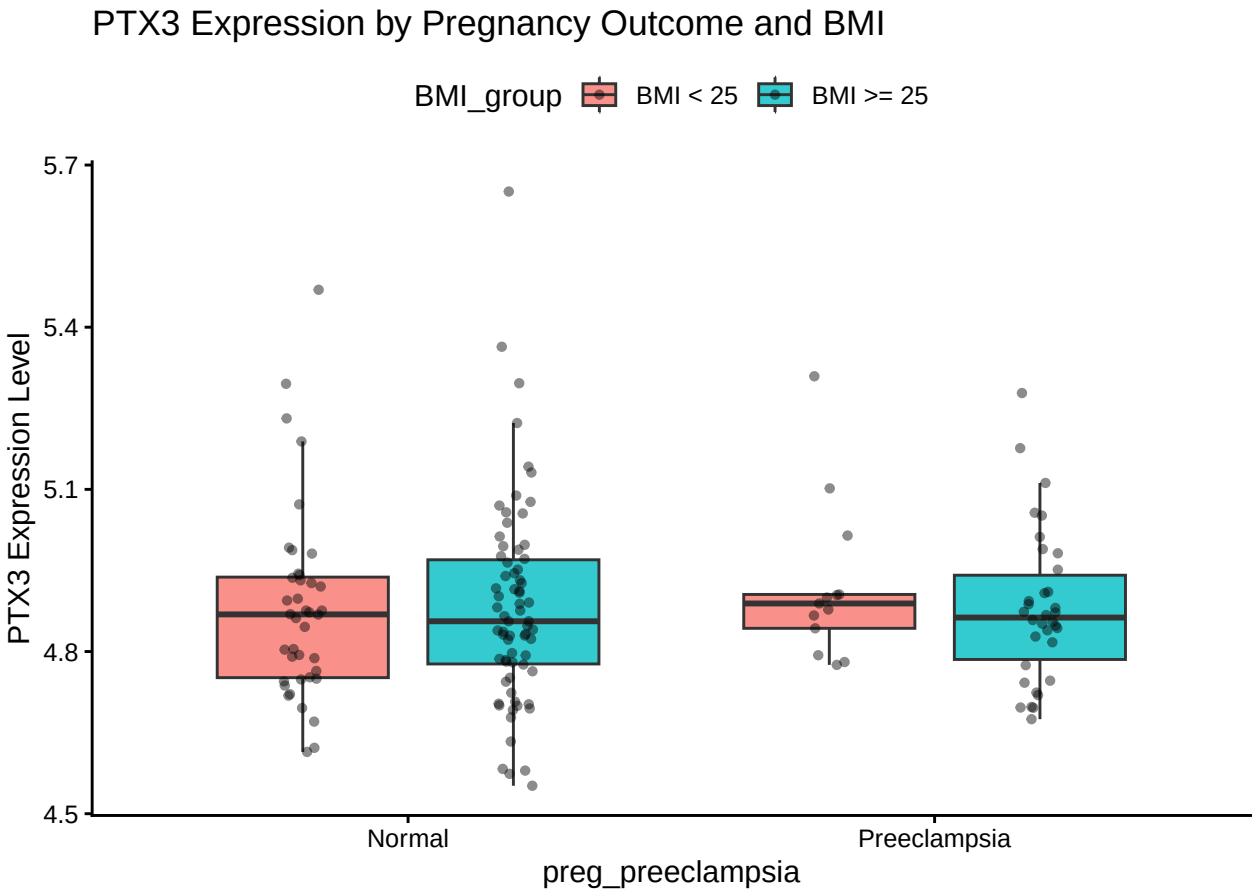
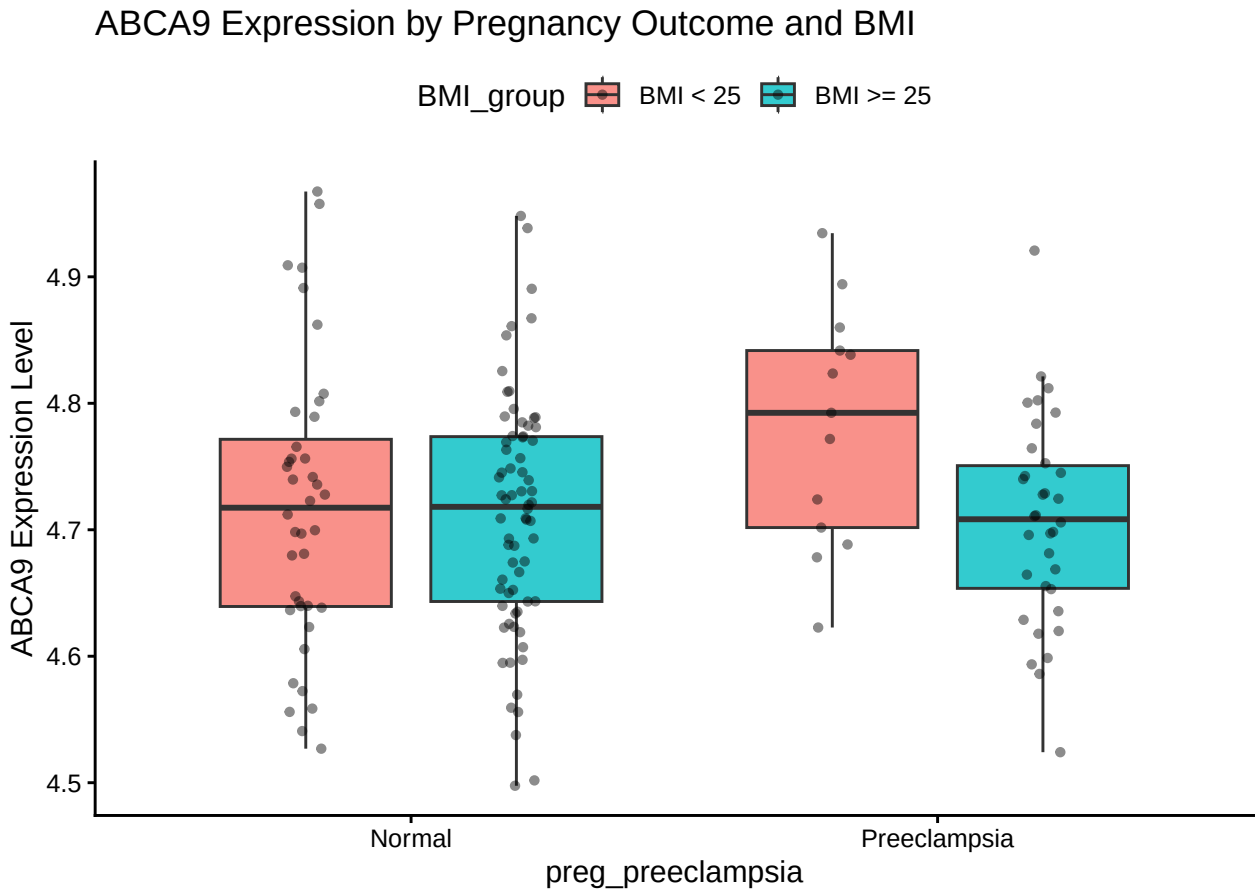
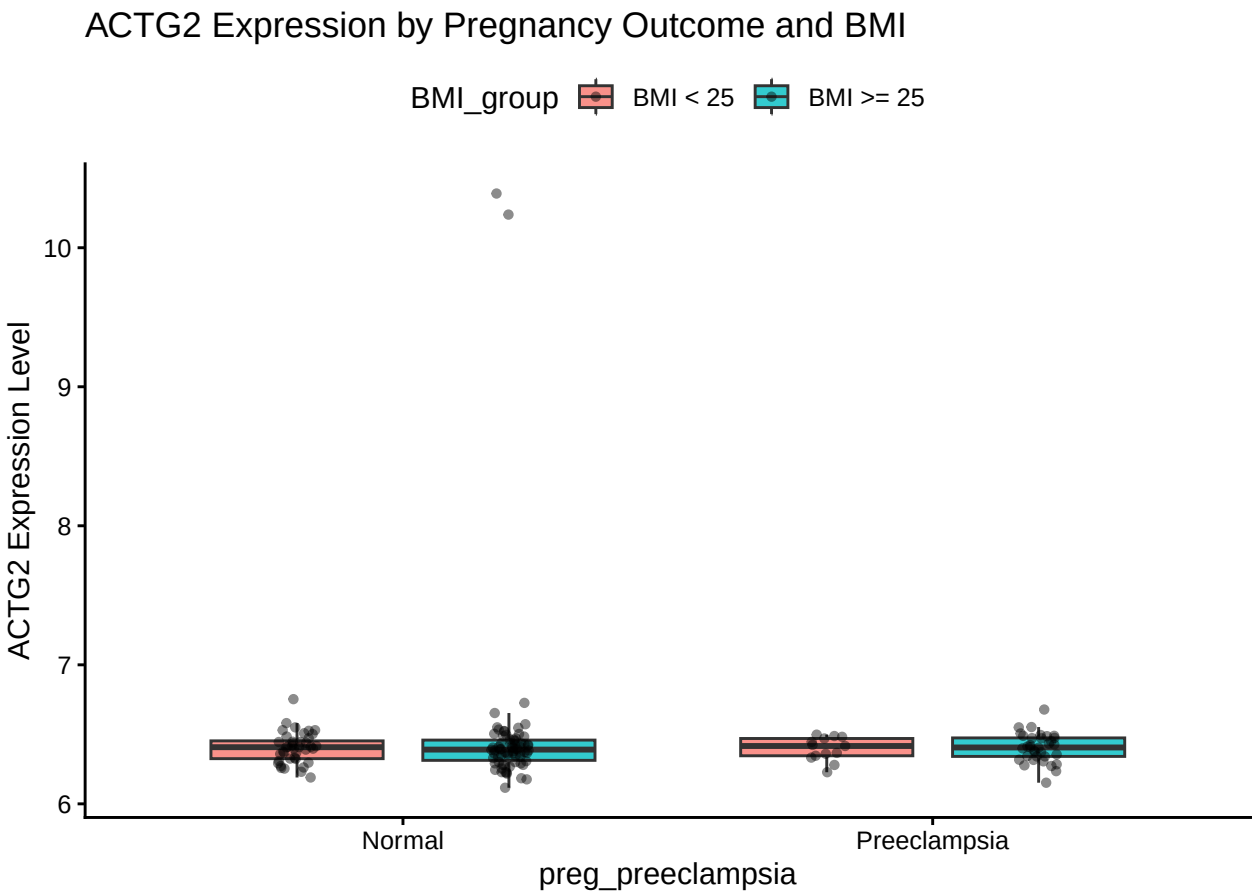


Figure Set 3: Gene Expression by Pregnancy Outcome and BMI

```
# Display boxplots with individual observations
for (gene in selected_genes) {
  print(all_plots[[gene]][["boxplot"]])
}
```


PTX3 Expression by Pregnancy Outcome and BMI
BMI_group
BMI < 25
BMI >= 25
PTX3 Expression Level
5.7
5.4
5.1
4.8
4.5
Normal
Preeclampsia
preg_preeclampsia


ABCA9 Expression by Pregnancy Outcome and BMI
BMI_group
BMI < 25
BMI >= 25
ABCA9 Expression Level
4.9
4.8
4.7
4.6
4.5
Normal
Preeclampsia
preg_preeclampsia



AI Use Disclosure

I used ChatGPT (OpenAI) as a supplementary resource to discuss the project approach, troubleshoot R code, receive guidance on GitHub setup and submission procedures, and improve the clarity and wording of written explanations. I reviewed the code, analyses, visualizations, and interpretations and take responsibility for the submitted work.